

A Simple System that Uses Duplicati for Backing Up Data

The open source based data backup system outlined here can be really useful for a small office home office setup. Learn how to set up Duplicati on Windows, back up data to a Linux server, schedule daily backups, secure them with encryption, restore files when needed, and receive email notifications.



Small office and home office (SOHO) environments often rely on a few computers and basic storage devices to manage essential data like client records, invoices, and project files. While the setup is simple, the data is valuable and losing it can cause serious disruptions. Data loss can occur due to hardware failures, accidental deletion, or malware attacks. Yet, many SOHO users lack a proper backup system or depend on manual copies that are easy to overlook.

So let's learn how to build a secure and automated backup system using Duplicati, an open source tool that supports encryption, scheduling, and both local and network storage.

System overview

To demonstrate the backup system, we use a typical SOHO arrangement involving two devices on the same local network:

- A Windows laptop, used for regular office tasks such as document editing, billing, and project handling.
- An Ubuntu virtual machine hosted locally using VirtualBox, which serves as the backup destination.
- Duplicati, installed on the Windows machine to manage the backup process.

The Ubuntu virtual machine is prepared to receive secure file transfers using the SFTP protocol. The following steps were followed to configure the Linux server as a backup target.

Install SSH server: To allow secure access, the SSH service is installed using the following steps:

```
$ sudo apt update && sudo apt install  
openssh-server
```

Create a backup user: A dedicated user named 'backupuser' is created to handle incoming backup data:

```
$ sudo adduser backupuser
```

A password is set during the process to authenticate incoming connections.

Create backup directory: A directory is created for storing backup files, and ownership is assigned to the new user:

```
$ sudo mkdir -p /home/backupuser/  
duplicati-backups  
$ sudo chown backupuser:backupuser /  
home/backupuser/duplicati-backups
```

Confirm SSH is running: The status of the SSH server is checked to ensure it is active and listening on port 22:

```
$ sudo systemctl status ssh
```

Find the server's IP address: The IP address of the Ubuntu machine is required to configure the Duplicati destination. It can be found using: `$ ip a`

For commands that begin with `sudo`, you will be prompted to enter your system password to authorise the action.

In our setup, the IP address is 192.168.1.6, which will be used by Duplicati to connect to the backup server using SFTP. This completes the server-side setup. The system is now ready to accept backups securely from the Windows machine. Now let's install and configure Duplicati to connect to this server and begin the backup process.

Installing Duplicati

With the backup server ready, the next step is to install Duplicati on the